



FINAL EXAM BRIDGING SEMESTER I, SESSION 2020/2021

COURSE CODE	:	IFC 1024
COURSE NAME	:	CHEMISTRY
PROGRAMME	:	BRIDGING SPACE
DURATION	:	3 HOURS
DATE	:	09 FEBRUARY 2021

INSTRUCTION TO CANDIDATES:

- 1.DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DOSO.
2. ANSWER ALL QUESTIONS.

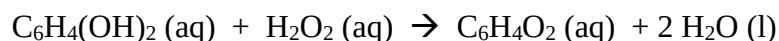
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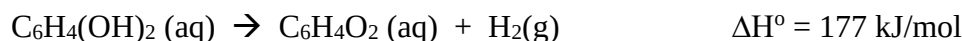
This booklet consists of (15) printed pages including this page.

QUESTION 1 (15 MARKS)

- a) A bombardier beetle (*Brachinus*) adopt a less passive defense mechanism. When threatened, the beetle squeezes some fluid from the inner compartment into the outer compartment, where in the presence of the enzymes, an exothermic reaction takes place :



Calculate the enthalpy of reaction from the given equations below :



$$q = C_p \Delta T \quad (5 \text{ marks})$$

- b) A quantity of 1.435 g of naphthalene, C_{10}H_8 , a pungent-smelling substance used in moth repellents, was burned in a constant-volume bomb calorimeter. The temperature rose from 20.28°C to 25.95°C. If the heat capacity of the bomb plus water was 10.17 kJ/°C, calculate the molar heat of combustion of naphthalene.

$$\Delta T = 5.67^\circ\text{C}, \quad q = 57.66 \text{ kJ} \quad \Delta H = \frac{-57.66 \text{ kJ}}{\frac{1.435}{128}} = -5143.2 \text{ kJ/mol} \quad (4 \text{ marks})$$

- c) A copper pellet having a mass of 26.47 g at 89.98°C was placed in a constant-pressure calorimeter containing 100.0 mL of water. The water temperature increase from 22.50°C to 23.17°C. Assuming that no heat is lost to the surroundings, calculate the specific heat of the copper pellet. [$S(\text{H}_2\text{O}) = 4.184 \text{ J/g} \cdot ^\circ\text{C}$ and $d(\text{H}_2\text{O}) = 1 \text{ g/mL}$]

$$m_s \Delta T = m_w \Delta T \quad S = 0.15 \text{ J/g} \cdot ^\circ\text{C} \quad (4 \text{ marks})$$

$$26.47(5)(89.98 - 23.17) = (100)(4.184)(0.67)$$

- d) Determine whether each process is exothermic or endothermic :

- Hand sanitizer evaporating from skin *endo*
- Hot pack used as muscle pain reliever *exo*

(1+1 marks)

QUESTION 2 (10 MARKS)

a) Given a reaction $\text{:NH}_4^+(\text{aq}) + \text{NO}_2^-(\text{aq}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

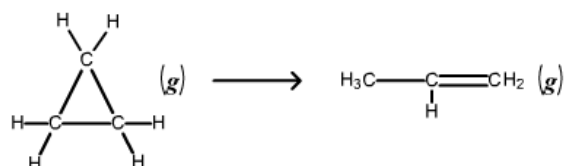
Experiment	NH_4^+	NO_2^-	Rate
1	0.35	0.35	8.0×10^{-4}
2	0.70	0.35	3.2×10^{-3}
3	0.70	0.70	6.4×10^{-3}

Based on the reaction and table above :

- Determine the order of the reaction for each reactant and the overall order of reaction
- Calculate the rate constant, k

(7+1 marks)

b) Cyclopropane (C_3H_6) is the smallest cyclic hydrocarbon. It is thermally unstable and rearranges to propene at 1000°C via the following first order reaction.



If the rate constant is 8.2 s^{-1} , calculate the half-life of the reaction.

(2 marks)

$$2) (a) (i) \frac{3.2 \times 10^{-3}}{8.0 \times 10^{-4}} = \frac{k(0.70)^x \cancel{(0.35)^y}}{k(0.35)^x \cancel{(0.35)^y}} \quad \frac{6.4 \times 10^{-3}}{3.2 \times 10^{-3}} = \frac{k \cancel{(0.70)^x} (0.70)^y}{k \cancel{(0.70)^x} (0.35)^y}$$

$$4 = 2^x, x = 2$$

$$2 = 2^y, y = 1$$

$$R = k[\text{NH}_4^+]^2[\text{NO}_2^-]$$

$$(ii) 8.0 \times 10^{-4} = k(0.35)^2(0.35)$$

$$k = 0.0187 \text{ M}^{-2} \text{ s}^{-1}$$

$$(b) k = 8.2, t_{1/2} = \frac{0.693}{k} = 0.0845 \text{ s}$$

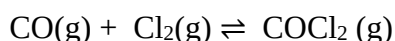
QUESTION 3 (15 MARKS)

- a) Consider the following equilibrium systems. Predict the direction of the net reaction in each case as a result of decreasing the pressure (increasing the volume) on the system at constant temperature.

- i. $2 \text{PbS (s)} + 3 \text{O}_2 \text{(g)} \rightleftharpoons 2 \text{PbO (s)} + 2 \text{SO}_2 \text{(g)}$
- ii. $\text{PCl}_5 \text{(g)} \rightleftharpoons \text{PCl}_3 \text{(g)} + \text{Cl}_2 \text{(g)}$
- iii. $\text{H}_2 \text{(g)} + \text{CO}_2 \text{(g)} \rightleftharpoons \text{H}_2\text{O (g)} + \text{CO (g)}$

(1+1+1 marks)

- b) Carbonyl chloride (COCl_2) also called phosgene, was used in World War I was a poisonous gas. The equilibrium concentrations for the reaction between carbon monoxide (CO) and molecular chlorine (Cl_2) to form COCl_2 is given below :



At equilibrium with 77°C , there are $[\text{CO}] = 2.1 \times 10^{-3} \text{ M}$, $[\text{Cl}_2] = 0.065 \text{ M}$, and $[\text{COCl}_2] = 0.42 \text{ M}$. Calculate the equilibrium constant, K_c and then K_p

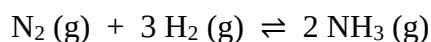
(4 marks)

- c) The equilibrium constant, K_p for the decomposition of carbonic acid (H_2CO_3) in soft drinks is found to be 1.05 at 250°C . If the equilibrium partial pressure of H_2CO_3 and CO_2 are 0.875 atm and 0.463 atm, respectively, what is the equilibrium partial pressure of H_2O at 250°C ?



(3 marks)

- d) At the start of a reaction, there a 0.429 mol N_2 , 0.0123mol H_2 , and 6.24×10^{-3} mol NH_3 in a 5.30 L reaction vessel at 375°C . If the equilibrium constant, K_c for the reaction is 1.2 at this temperature, decide whether the system is at equilibrium. If it's not, predict which way the equilibrium will proceed.



(5 marks)

- 3) (a) (i) shift to left
 (ii) shift to right
 (iii) no shift

(b) $\text{CO(g)} + \text{Cl}_2\text{(g)} \rightleftharpoons \text{CoCl}_2\text{(g)}$, $T = 350\text{K}$, $[\text{Co}] = 2.1 \times 10^{-3}\text{M}$, $[\text{Cl}_2] = 0.065\text{M}$, $[\text{CoCl}_2] = 0.42\text{M}$

$$K_c = \frac{[\text{CoCl}_2]}{[\text{Co}][\text{Cl}_2]} = \frac{0.42}{0.065 \times 2.1 \times 10^{-3}} = 3077$$

$$K_p = K_c RT^{\Delta n}$$

$$= 3077 \times (0.0821 \times 350)^{1-2}$$

$$= 107.08$$

(c) $K_c = \frac{(\text{CO}_2)(\text{H}_2\text{O})}{(\text{H}_2\text{CO}_3)} = 1.05$, $T = 523\text{K}$

$P(\text{H}_2\text{CO}_3) = 0.875\text{ atm}$, $P(\text{CO}_2) = 0.463\text{ atm}$, $P(\text{H}_2\text{O}) = ?$

$$\frac{0.463 \times P(\text{H}_2\text{O})}{0.875} = 1.05$$

$$P(\text{H}_2\text{O}) = 1.984\text{ atm}$$

(d) $[\text{N}_2]_i = 0.429\text{ mol}$
 $[\text{H}_2]_i = 0.0123\text{ mol}$
 $[\text{NH}_3]_i = 6.24 \times 10^{-3}\text{ mol}$
 volume : 5.30 L
 temp: 648 K
 $K_c = 1.2$

$$Q_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

$$= \frac{(6.24 \times 10^{-3})^2}{0.429(0.0123)^3}$$

$$= 0.6$$

$$Q_c < K_c$$

$$0.6 < 1.2$$

$\therefore$ the system has too much reactant,
 hence the system will shift to right

QUESTION 4 (15 MARKS)

- a) Calculate the concentration of 100 mL hydrochloric acid, HCl, that has a pH of 0.75
(2 marks)
- b) Consider the mixing of sodium hypobromite (NaBrO) into 2.00 L of 0.25 M hypobromous acid (HBrO) to form a buffer solution (K_a of HBrO = 2.3×10^{-9})?
- Assuming that no volume change occurs when the NaBrO is added, Calculate the number of moles of NaBrO need to be added into the solution to form a buffer solution with pH of 8.20
 - Calculate the final pH if 10 mL of 0.5 M NaOH is added into the solution.
(4+5 marks)
- c) A 0.100 M solution of chloroacetic acid (ClCH_2COOH) has a K_a value of 1.3×10^{-3} . Calculate the percentage ionization of the acid.
(4 marks)

4) (a) $\text{pH} = -\log [\text{H}^+]$
 $0.75 = -\log [\text{H}^+]$
 $[\text{H}^+] = 10^{-0.75}$
 $= 0.177 \text{ M}$

(b) (i) weak acid: HBrO

mol of $\text{HBrO} = 2 \times 0.25 = 0.5 \text{ mol}$

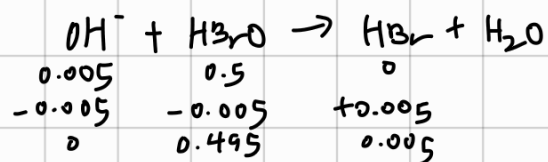
conjugate base: BrO^-

salt of conjugate base: NaBrO

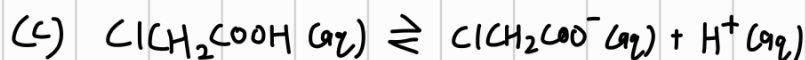
$\text{pH} = \text{pK}_a + \log \frac{[\text{c. base}]}{[\text{acid}]}$
 $8.2 = -\log(2.3 \times 10^{-9}) + \log \frac{[x]}{0.5}$
 $[x] = 0.1823 \text{ mol}$

(ii) mol of NaOH added = $0.01 \times 0.5 = 0.005 \text{ mol}$

mol of $\text{OH}^- \equiv \text{mol of NaOH} = 0.005 \text{ mol}$



$\text{pH} = -\log(2.3 \times 10^{-9}) + \log\left(\frac{0.005}{0.495}\right)$
 $= 6.64$



0.1	0	0
-x	+x	+x
0.1-x	x	x

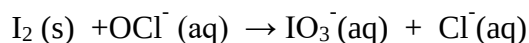
$1.3 \times 10^{-3} = \frac{x^2}{0.1}$

$x = 0.0114$

% ionization = $\frac{0.0114}{0.1} \times 100$
 $= 11.4 \%$

QUESTION 5 (15 MARKS)

a) Complete and balance the following equation in acidic solution:

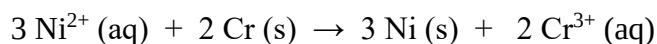


(5 marks)

b) Give TWO functions of salt bridge in a voltaic cell setup

(2 marks)

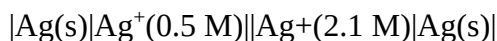
c) Consider the redox reaction below:



- Using standard reduction potentials, calculate the standard EMF for the reaction
- Write the half reaction equations
- Write the short-hand notation for the voltaic cell above

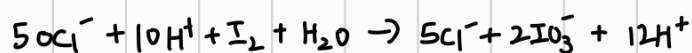
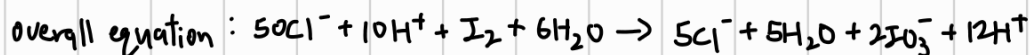
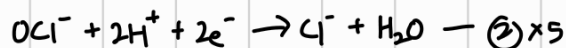
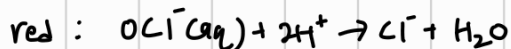
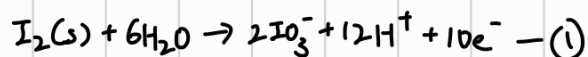
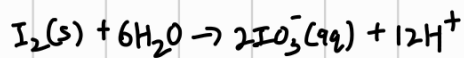
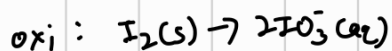
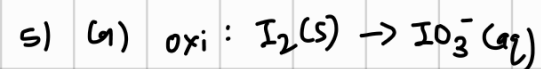
(1+2+1 marks)

d) Consider a concentration cell (voltaic cell) at temperature of 25°C. The setup is as below



- Which silver solution should be on the anode and cathode?
- Calculate the voltage produced by the cell?

(1+3 marks)



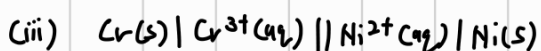
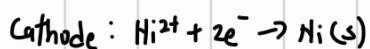
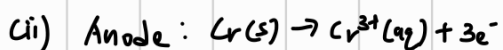
(b) - to remain the solution electrically neutral

- to allow flow of electrons



(c) (i) $E^\circ_{cell} = -0.25 - (-0.74)$

$$= +0.49V$$



(d) (i) Anode: $0.5M$

Cathode: $2.1M$

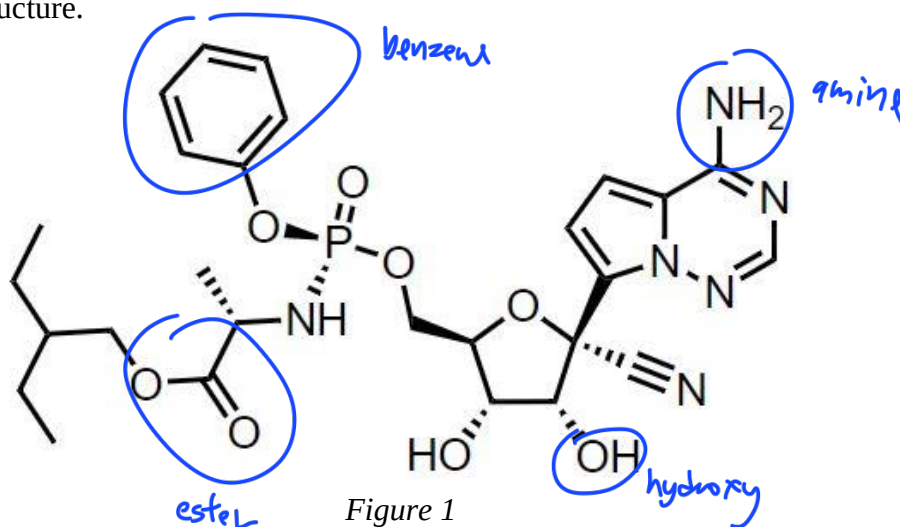
(ii) $E_{cell} = E^\circ_{cell} - \frac{0.0592}{n} \log Q$

$$= 0 - \frac{0.0592}{1} \log \frac{0.5}{2.1}$$

$$= 0.037V$$

QUESTION 6 (10 MARKS)

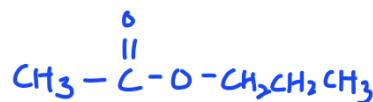
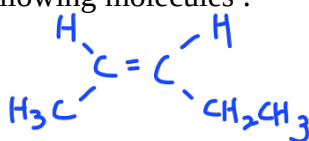
- a) Figure 1 shows a structural formula of remdesivir, an antiviral drug which in the past was touted as a possible remedy for Covid-19. Draw and list FOUR functional groups from the structure.



(4 marks)

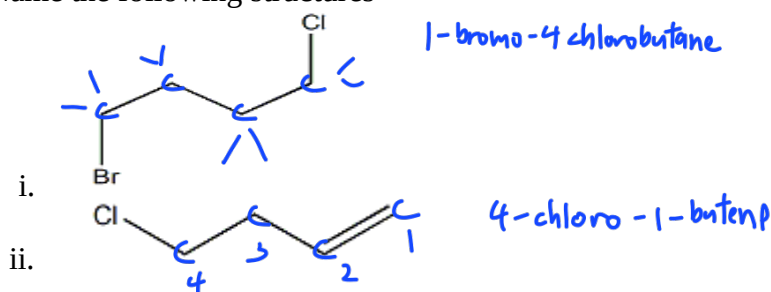
- b) Draw the structure of the following molecules :

- Cis-2-pentene
- Propyl ethanoate



(1+1 marks)

- c) Name the following structures



(1+1 marks)

- d) Draw TWO (2) possible structures of C_4H_8 .



(2 marks)

List of Selected Constant Values

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

Unit and Conversion Factor

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

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